Project: Worked examples
Title: Duct sizing
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Sizes each duct section by the equal-friction method (a target friction rate) or the velocity method (a target velocity), rounds to standard sizes, and reports velocity, friction rate, friction and fitting losses per section and the total pressure drop along the index run, with warnings against user velocity and pressure-drop limits.

Round: D from the target (friction: Colebrook with the air properties at T; velocity: D = √(4Q/πv)), rounded up to the next standard size. Rectangular: equivalent diameter D_e = 1.30 (ab)^0.625/(a + b)^0.25 (equal friction and capacity); flat oval: D_e = 1.55 A^0.625/P^0.25 with A = πb²/4 + b(a − b), P = πb + 2(a − b). Friction rate Δp/L = f ρ v_e²/(2D_e) with v_e the velocity in the equivalent round duct; fitting loss ΣC ρ v²/2 at the actual velocity; totals summed over the index-run sections.

Ref: ASHRAE Handbook — Fundamentals Ch. 21 — Duct design (friction chart basis, equivalent diameters, loss coefficients)
Ref: CIBSE Guide C Section 4 — Flow of fluids in pipes and ducts
Ref: BS EN 1505 — Ventilation for buildings: sheet metal air ducts and fittings with rectangular cross-section, dimensions
Ref: BS EN 1506 — Ventilation for buildings: sheet metal air ducts and fittings with circular cross-section, dimensions

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
method	Sizing method	velocity		assumption: Velocity method for an extract system.	any
dp_target	Target friction rate (equal friction)	1	Pa/m	default	0.3 … 5 Pa/m
v_target	Target velocity (velocity method)	10	m/s	spec: Dust-carrying extract: 10 m/s minimum transport velocity, SPEC-M-09	2 … 20 m/s
T	Air temperature	25	degC	assumption: Workshop air 25 °C.	-20 … 60 degC
p_atm	Barometric pressure	100.5	kPa	assumption: Site 70 m AOD, ≈ 100.5 kPa.	80 … 105 kPa
eps	Duct roughness	0.15	mm	assumption: Spiral-wound galvanised duct with joints.	0.03 … 3 mm
sections	Duct sections	3 row(s)		drawing: DRG-M-310 rev A	any
v_limit	Velocity limit (warning and check)	15	m/s	spec: SPEC-M-09 noise limit	2 … 30 m/s
dp_limit	Total pressure-drop limit (check)	600	Pa	spec: Extract fan external static 600 Pa	50 … 2000 Pa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
round_to	Round sizes up to a multiple of	50	mm	spec: BS EN 1506 preferred sizes	1 … 100 mm

3. Assumptions

A1. Air is dry at the stated temperature and pressure (ρ = p/(287.055 T)); viscosity by Sutherland's law. (Standard air properties.)
A2. Rectangular and flat-oval ducts are sized through the equivalent diameter for equal friction and capacity. (Huebscher / Heyt–Diaz relations as given in ASHRAE Fundamentals Ch. 21.)
A3. Fitting loss coefficients C are user-supplied (manufacturer or ASHRAE/CIBSE data, not embedded) and refer to the section's velocity pressure. (Non-negotiable: no standard tables embedded.)
A4. method (Sizing method) is taken as velocity. (Velocity method for an extract system.)
A5. T (Air temperature) is taken as 25 degC. (Workshop air 25 °C.)
A6. p_atm (Barometric pressure) is taken as 100.5 kPa. (Site 70 m AOD, ≈ 100.5 kPa.)
A7. eps (Duct roughness) is taken as 0.15 mm. (Spiral-wound galvanised duct with joints.)

4. Working

[10] rho = p / (287.055 T)
= 100500 Pa / (287.055 × 298.15 K)
= 1.1743 kg/m³
Air density

[11] mu = Sutherland's law
= T = 298.15 K
= 18.371 uPa·s
Air dynamic viscosity

[12] D_s1 = hood branch: diameter
= target 211.1 mm rounded up to 50 mm
= 250 mm
Diameter, hood branch

[13] v_s1 = hood branch: Q / A
= 0.35 / 0.049087
= 7.1301 m/s
Velocity, hood branch

[14] Re_s1 = hood branch: ρ v_e D_e / μ (v_e = 4Q/πD_e²)
= 1.1743 × 7.1301 × 0.25 / 0.000018371
= 113940
Reynolds number, hood branch

[15] f_s1 = hood branch: friction factor
= Colebrook–White
= 0.02041
Friction factor, hood branch

[16] dpL_s1 = hood branch: f ρ v_e² / (2 D_e)
= 0.02041 × 1.1743 × 7.1301² / (2 × 0.25)
= 2.4369 Pa/m
Friction rate, hood branch

[17] dp_f_s1 = hood branch: (Δp/L) L
= 2.4369 × 6
= 14.622 Pa
Friction loss, hood branch

[18] dp_fit_s1 = hood branch: ΣC ρ v²/2
= 1.5 × 29.849
= 44.774 Pa
Fitting loss, hood branch

[19] dp_s1 = hood branch: friction + fittings + equipment
= 14.622 + 44.774 + 120
= 179.4 Pa
Section loss, hood branch (index run)
[20] D_s2 = main: diameter
= target 365.64 mm rounded up to 50 mm
= 400 mm
Diameter, main
[21] v_s2 = main: Q / A
= 1.05 / 0.12566
= 8.3556 m/s
Velocity, main
[22] Re_s2 = main: ρ v_e D_e / μ (v_e = 4Q/πD_e²)
= 1.1743 × 8.3556 × 0.4 / 0.000018371
= 213630
Reynolds number, main
[23] f_s2 = main: friction factor
= Colebrook–White
= 0.018028
Friction factor, main
[24] dpL_s2 = main: f ρ v_e² / (2 D_e)
= 0.018028 × 1.1743 × 8.3556² / (2 × 0.4)
= 1.8475 Pa/m
Friction rate, main
[25] dp_f_s2 = main: (Δp/L) L
= 1.8475 × 20
= 36.95 Pa
Friction loss, main
[26] dp_fit_s2 = main: ΣC ρ v²/2
= 0.6 × 40.992
= 24.595 Pa
Fitting loss, main
[27] dp_s2 = main: friction + fittings + equipment
= 36.95 + 24.595 + 0
= 61.545 Pa
Section loss, main (index run)
[28] D_s3 = stack: diameter
= target 365.64 mm rounded up to 50 mm
= 400 mm
Diameter, stack
[29] v_s3 = stack: Q / A
= 1.05 / 0.12566
= 8.3556 m/s
Velocity, stack
[30] Re_s3 = stack: ρ v_e D_e / μ (v_e = 4Q/πD_e²)
= 1.1743 × 8.3556 × 0.4 / 0.000018371
= 213630
Reynolds number, stack
[31] f_s3 = stack: friction factor
= Colebrook–White
= 0.018028
Friction factor, stack

[32] dpL_s3 = stack: f ⋅ v_e² / (2 D_e)
= 0.018028 × 1.1743 × 8.3556² / (2 × 0.4)
= 1.8475 Pa/m
Friction rate, stack
[33] dp_f_s3 = stack: (Δp/L) L
= 1.8475 × 8
= 14.78 Pa
Friction loss, stack
[34] dp_fit_s3 = stack: ΣC ⋅ v²/2
= 1 × 40.992
= 40.992 Pa
Fitting loss, stack
[35] dp_s3 = stack: friction + fittings + equipment
= 14.78 + 40.992 + 30
= 85.772 Pa
Section loss, stack (index run)
[36] dp_index = Σ section losses on the index run
= 3 section(s)
= 326.71 Pa
Index-run pressure drop
[37] dp_all = Σ section losses, all sections
= 3 section(s)
= 326.71 Pa
Sum over every section (not a path)
[38] v_peak = highest section velocity
=
= 8.3556 m/s
Highest velocity
[39] u_dp = dp_index / dp_limit
= 326.71 / 600
= 0.54452
Pressure-drop utilisation
[40] u_v = v_peak / v_limit
= 8.3556 / 15
= 0.55704
Velocity utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
D_s1	250 mm	D, hood branch
v_s1	7.1301 m/s	v, hood branch
Re_s1	113940	Re, hood branch
f_s1	0.02041	f, hood branch
dpL_s1	2.4369 Pa/m	Δp/L, hood branch
dp_f_s1	14.622 Pa	Friction loss, hood branch
dp_fit_s1	44.774 Pa	Fitting loss, hood branch
dp_s1	179.4 Pa	Section loss, hood branch
D_s2	400 mm	D, main
v_s2	8.3556 m/s	v, main

SYMBOL	VALUE	DESCRIPTION
Re_s2	213630	Re, main
f_s2	0.018028	f, main
dpL_s2	1.8475 Pa/m	Δp/L, main
dp_f_s2	36.95 Pa	Friction loss, main
dp_fit_s2	24.595 Pa	Fitting loss, main
dp_s2	61.545 Pa	Section loss, main
D_s3	400 mm	D, stack
v_s3	8.3556 m/s	v, stack
Re_s3	213630	Re, stack
f_s3	0.018028	f, stack
dpL_s3	1.8475 Pa/m	Δp/L, stack
dp_f_s3	14.78 Pa	Friction loss, stack
dp_fit_s3	40.992 Pa	Fitting loss, stack
dp_s3	85.772 Pa	Section loss, stack
dp_index	326.71 Pa	Index-run Δp
dp_all	326.71 Pa	Σ all sections
v_peak	8.3556 m/s	Highest velocity

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Velocity ≤ limit	0.55704	1	0.557	
PASS	Index-run Δp ≤ limit	0.54452	1	0.545	

7. Applicability

No warnings.

8. Verification

Template bs.duct-sizing@1.0.0 Content hash 1d090889cef32044 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0